

3.3.2 Functional and chemical properties of food

3.3.2.1 Proteins

Content	Students must know and understand	Suggested application and food preparation skills
- protein denaturation - protein coagulation - gluten formation - foam formation.	- the scientific principles underlying these processes when preparing and cooking food - the working characteristics, functional and chemical properties of proteins.	- Demonstrate how acids denature protein and marinades add flavour and moisture when preparing vegetables, meat, fish and alternatives (S9). - Setting of egg mixtures eg in quiche (S12). - Gluten formation – pasta making using a pasta machine, bread making using a bread machine (S5 and S10). - The use of marinades to tenderise and flavour meats and alternatives (S9). - Whisking eggs to produce a gas-in-liquid foam eg whisked sponge.

3.3.2.2 Carbohydrates

Content	Students must know and understand	Suggested application and food preparation skills
- gelatinisation - dextrinisation - caramelisation.	- the scientific principles underlying these processes when preparing and cooking food - the working characteristics, functional and chemical properties of carbohydrates.	- Make a blended white sauce showing starch gelatinisation such as either a roux or all-in-one blended sauce, infused sauce, velouté or béchamel to demonstrate how liquid/starch ratios affect viscosity (S8). - Demonstrate how conduction and convection work to cook the sauce and the need for agitation. - Caramelisation of vegetables (S6). - Dextrinisation eg browning of bread when baking (S4).

3.3.2.3 Fats and oils

Content	Students must know and understand	Suggested application and food preparation skills
- shortening - aeration - plasticity - emulsification.	- the scientific principles underlying these processes when preparing and cooking food - the working characteristics, functional and chemical properties of fats and oils.	- Use of fats/oils to demonstrate these processes. - Shortening and plasticity, eg pastry making (S10). - Aeration eg using the creaming method with a food mixer for a cake (S1, S4, S5 and S11). - Make an emulsion sauce such as a salad dressing, mayonnaise or hollandaise (S8).

3.3.2.4 Fruit and Vegetables

Content	Students must know and understand	Suggested application and food preparation skills
- enzymic browning - oxidation.	the scientific principles underlying these processes when preparing and cooking food.	- When preparing fresh fruits such as apples and pears, preventing enzymic browning by using lemon juice (S2) and (S3). - Oxidation eg preventing water soluble vitamin loss when preparing and cooking vegetables (S3) and (S6).

3.3.2.5 Raising agents

Content	Students must know and understand	Suggested application and food preparation skills
- chemical (baking powder, bicarbonate of soda, self-raising flours which produce carbon dioxide) - mechanical (whisking, beating, folding, sieving, creaming and rubbing in – all incorporate air into the mixture) - steam is produced when the water in any moist mixture reaches boiling point - biological (yeast).	- the scientific principles underlying these processes when preparing and cooking food - the working characteristics, functional and chemical properties of raising agents.	- Using chemical raising agents such as self-raising flour and baking powder (S11). - Use steam in a mixture to raise choux pastry or batter. - Use egg as a raising agent to: - create a gas-in-liquid foam - whisk egg whites - whisking savoury roulade. - Yeast in bread making.